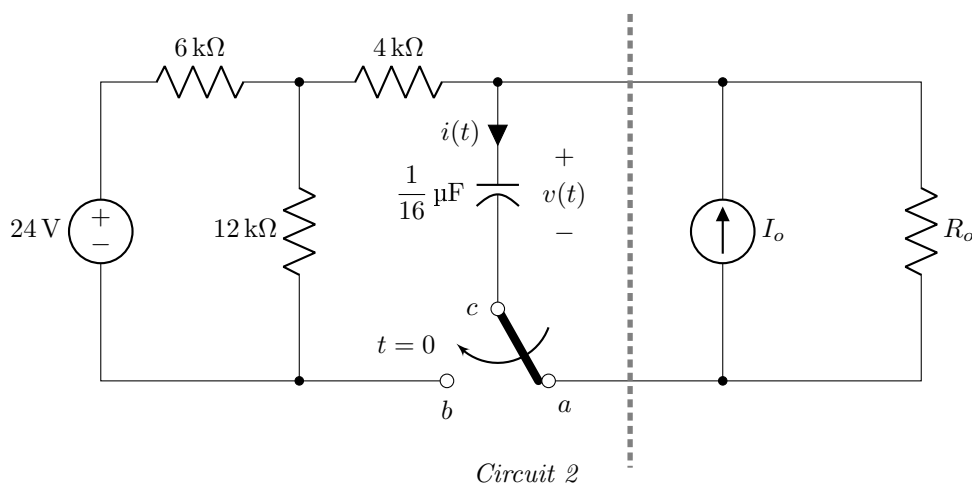
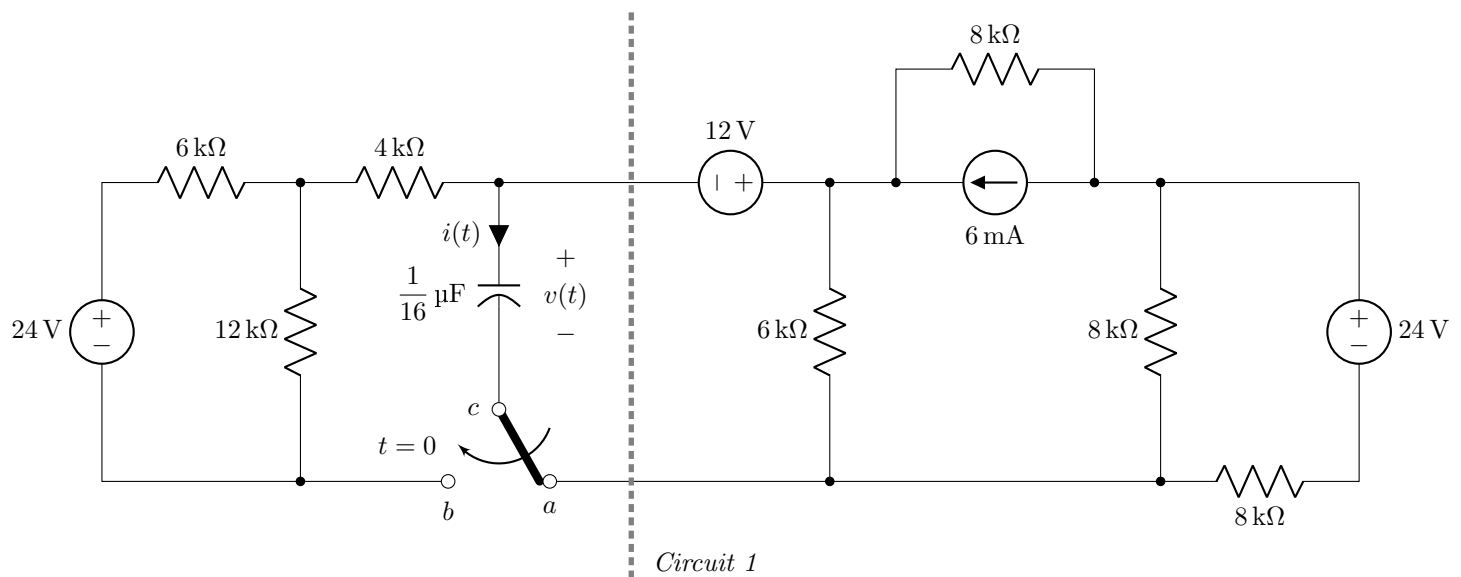


- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ All **3 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 3

[CO3] [20 marks]



Analyze the circuits shown above to answer the following questions:

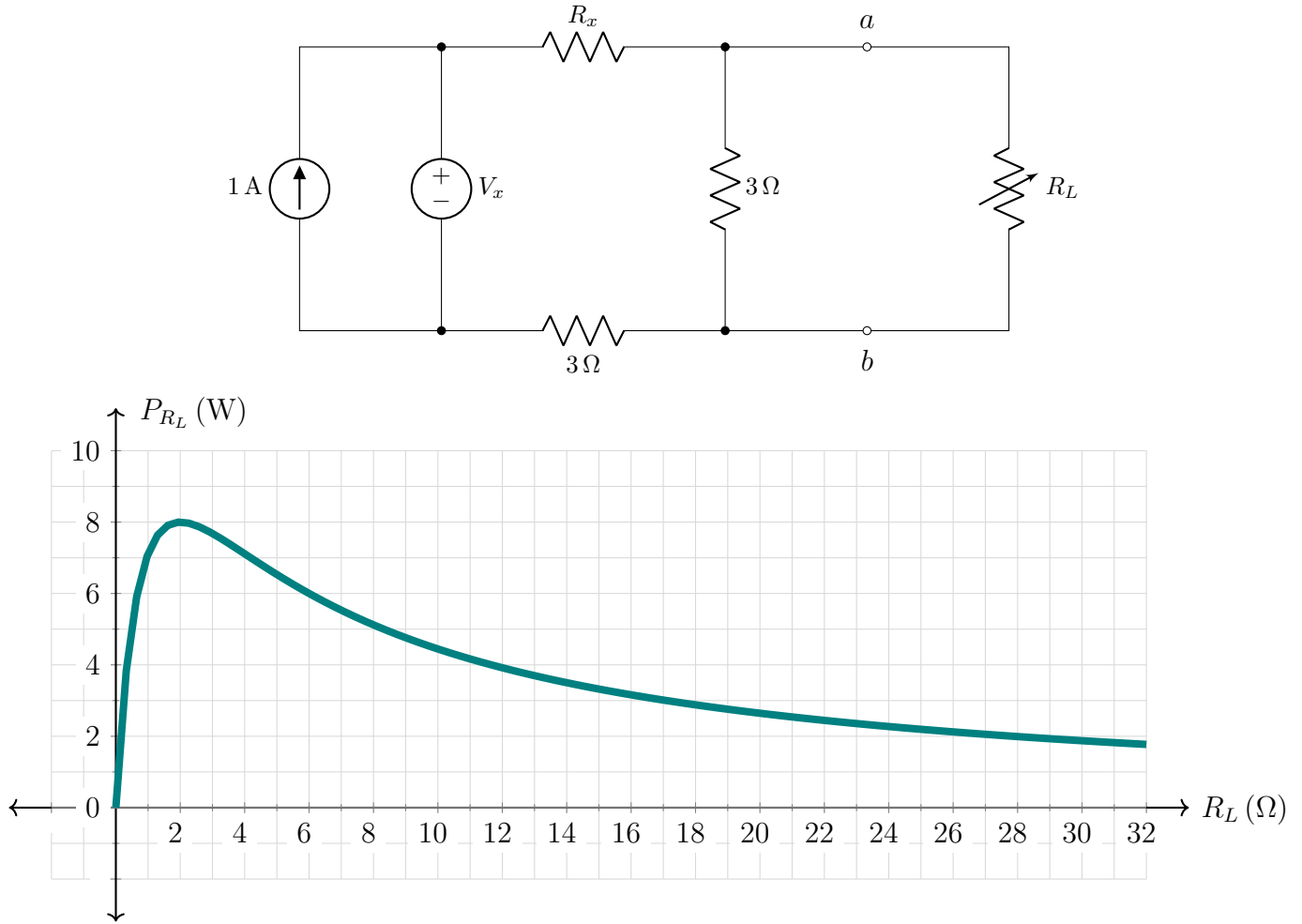
- (a) [8 marks] Reduce the portion of *Circuit 1* to the right of the vertical dashed line[†] so that it takes the form of *Circuit 2*. Hence, determine the values of I_o and R_o .
- (b) [9 marks] Determine the voltage $v(t)$ across the capacitor as a function of time t , for $t > 0$.
- (c) [3 marks] Determine the capacitor current $i(t)$ as a function of time, for $t > 0$.

[†]The vertical dashed line indicates a boundary for circuit reduction and does not represent any circuit element or physical connection.

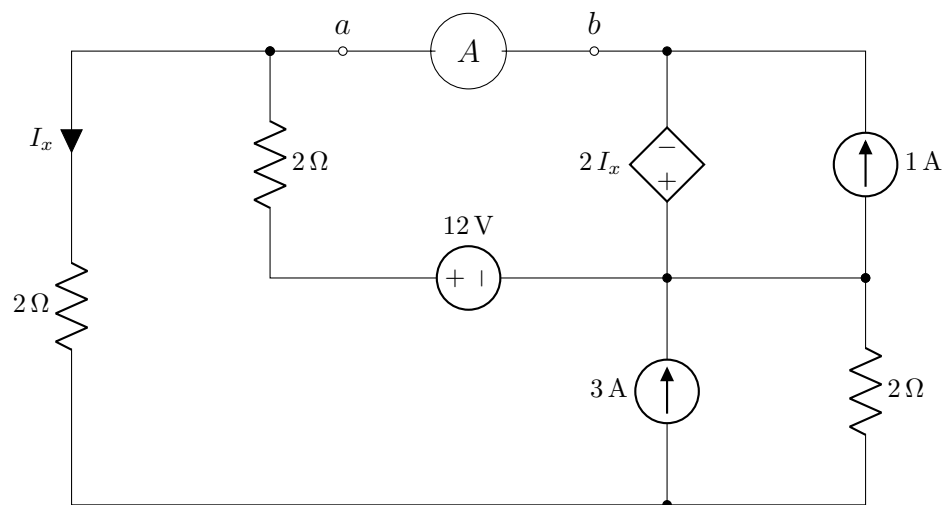
■ Question 2 of 3

[CO2] [23 marks]

(a) In the following circuit, the power delivered to the resistor R_L is plotted below as R_L is varied.



- [4 marks] Determine the unknown resistance R_x in the circuit.
 - [4 marks] Determine the unknown voltage V_x .
- (b) The ammeter A in the following circuit is used to measure the current flowing between terminals a and b .



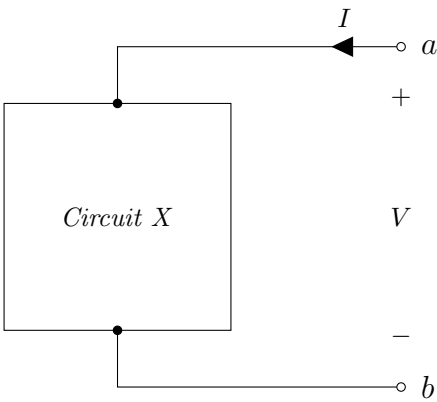
Apply Thevenin's Theorem or Norton's Theorem to answer the following questions:

- [12 marks] Determine the Thevenin or Norton equivalent circuit with respect to terminals a and b .
- [3 marks] If the ammeter can safely carry a maximum current of 1 A, use the equivalent circuit obtained in (i) to determine the minimum value of resistance of the ammeter required so that it operates safely without exceeding the maximum allowable current.

■ Question 3 of 3

[CO3] [12 marks]

(a) [4 marks] X is a linear two-terminal circuit connected between terminals a and b , as shown below. When tested, the circuit produced the voltage V and current I values listed in the adjacent table; however, two entries are missing. Calculate the missing values and complete the table.



V (volt)	2	6	10	
I (A)	-0.5	0.5		2.0

(b) [8 marks] Determine whether the following circuit is equivalent to $Circuit\ X$ in (a).

